

Midterm Exam 2

PhD Macroeconomics (Spring 2026)

Instructions. Show steps. You may explain in words where requested.

Question 1 (Asset pricing in an endowment economy)

[40 pts]

This question uses the notation from `AssetPricingEndowment.pdf`. In each period, the representative household receives fruit/dividend z . The household holds shares s of the fruit tree. The ex-dividend share price is p . Next period variables are denoted by primes: z' , p' , s' , c' . The household can also trade a one-period treasury bill: hold a' bought at price q today and paying 1 next period.

Budget constraint (with both assets):

$$c + ps' + qa' \leq (z + p)s + a. \quad (\text{BC})$$

Assume equilibrium goods market clearing implies $c = z$ and equilibrium share supply implies $s = 1$.

(a) **Euler equations and returns.** [16 pts]

(i) Show the FOC for the treasury bill implies

$$u'(c) q = \beta \mathbb{E}_{z'|z}[u'(c')]. \quad (\text{E-bond})$$

(ii) Show the FOC for shares implies

$$u'(c) p = \beta \mathbb{E}_{z'|z}[u'(c') (z' + p')]. \quad (\text{E-share})$$

(iii) Define the **gross returns** exactly as in the slides:

$$e(z, z') \equiv \frac{z' + p'}{p}, \quad R(z) \equiv \frac{1}{q}.$$

Rewrite (E-bond) and (E-share) as $1 = \mathbb{E}_{z'|z}[\text{SDF} \times \text{return}]$ using the slide's SDF.

(b) **Fundamental value (pricing by discounted dividends).** [14 pts]

Starting from (E-share) and using the transversality condition

$$\lim_{j \rightarrow \infty} \beta^j \mathbb{E}_t[u'(c_{t+j}) p_{t+j}] = 0, \quad (\text{TVC})$$

show that

$$u'(c_t) p_t = \mathbb{E}_t \left[\sum_{j=1}^{\infty} \beta^j u'(c_{t+j}) z_{t+j} \right]. \quad (\text{FV})$$

Then impose $c_t = z_t$ and rewrite p_t in terms of $\{z_{t+j}\}$ and marginal utilities (as in the slide).

(c) **Risk premium (covariance representation).**

[10 pts]

Using the covariance identity from the slides, show that

$$\frac{\mathbb{E}_{z'|z}[e(z, z')] - R(z)}{R(z)} = -\text{cov}_z \left(\underbrace{\beta \frac{u'(c')}{u'(c)}}_{\text{SDF}}, e(z, z') \right). \quad (\text{RP})$$

Give a one-sentence interpretation: when is the risk premium positive?

Solution. (a) (i) The FOC w.r.t. a' yields $u'(c)q = \beta \mathbb{E}_{z'|z}[u'(c')]$.

(ii) The FOC w.r.t. s' yields $u'(c)p = \beta \mathbb{E}_{z'|z}[u'(c')(z' + p')]$.

(iii) Define the SDF as in the slides:

$$\text{SDF} \equiv \beta \frac{u'(c')}{u'(c)}.$$

Divide (E-bond) by $u'(c)q$: $1 = \mathbb{E}_{z'|z}[\text{SDF} \cdot (1/q)] = \mathbb{E}_{z'|z}[\text{SDF} \cdot R(z)]$. Divide (E-share) by $u'(c)p$: $1 = \mathbb{E}_{z'|z}[\text{SDF} \cdot (z' + p')/p] = \mathbb{E}_{z'|z}[\text{SDF} \cdot e(z, z')]$.

(b) Iterating (E-share) forward gives

$$u'(c_t)p_t = \mathbb{E}_t \left[\sum_{j=1}^J \beta^j u'(c_{t+j}) z_{t+j} + \beta^J u'(c_{t+J}) p_{t+J} \right].$$

Under (TVC), the last term goes to 0 as $J \rightarrow \infty$, yielding (FV). With $c_t = z_t$, divide by $u'(z_t)$:

$$p_t = \mathbb{E}_t \left[\sum_{j=1}^{\infty} \beta^j \frac{u'(z_{t+j})}{u'(z_t)} z_{t+j} \right],$$

which matches the slide expression.

(c) Start from $1 = \mathbb{E}_{z'|z}[\text{SDF} \cdot e]$ and note $\mathbb{E}_{z'|z}[\text{SDF}] = 1/R(z)$ from the bond Euler equation. Use $\mathbb{E}[AB] = (A, B) + \mathbb{E}[A]\mathbb{E}[B]$:

$$1 = \mathbb{E}[\text{SDF} \cdot e] = \text{cov}_z(\text{SDF}, e) + \mathbb{E}[\text{SDF}]\mathbb{E}[e] = \text{cov}_z(\text{SDF}, e) + \frac{1}{R(z)}\mathbb{E}[e].$$

Rearrange:

$$\frac{\mathbb{E}[e] - R(z)}{R(z)} = -\text{cov}_z(\text{SDF}, e).$$

Risk premium is positive when the asset pays low returns in high-SDF (high marginal utility) states (“bad times”).

Question 2 (Asset pricing in a production economy) [30 pts]

This question uses the notation from `AssetPricingProduction.pdf`. A representative firm has labor-only technology

$$y = zn^\alpha, \quad \alpha \in (0, 1),$$

chooses labor n , pays wage w , and pays dividend $d = y - wn$. A representative household chooses (c, n, s') and has value function

$$V(s, z) = \max_{c \geq 0, s' \geq 0, n \geq 0} \left\{ \log c + \psi(1 - n) + \beta \mathbb{E}_{z'|z} [V(s', z')] \right\},$$

subject to

$$c + ps' \leq (d + p)s + wn.$$

In equilibrium: goods market clears $c = y$ and share market clears $s = 1$.

- (a) **Firm: wages and dividends.** [10 pts]

Show the firm FOC implies $w = \alpha zn^{\alpha-1}$. Show the dividend is $d = (1 - \alpha)y$.

- (b) **Household FOC for labor.** [8 pts]

Show the household FOC implies

$$\frac{w}{c} = \psi \quad \Rightarrow \quad w = \psi c. \quad (\text{L})$$

Use equilibrium $c = y$ and the firm wage equation to solve for

$$n = \frac{\alpha}{\psi} \quad \text{and} \quad y = z \left(\frac{\alpha}{\psi} \right)^\alpha.$$

- (c) **Share Euler equation and constant p/y .** [12 pts]

From the household FOC for s' , show:

$$\frac{1}{c} p = \beta \mathbb{E}_{z'|z} \left[\frac{1}{c'} (d' + p') \right]. \quad (\text{S})$$

Using $c = y$ and $d' = (1 - \alpha)y'$, show:

$$\frac{p}{y} = \beta(1 - \alpha) + \beta \mathbb{E}_{z'|z} \left[\frac{p'}{y'} \right].$$

Now guess $\frac{p}{y} \equiv \Lambda$ is constant and solve for Λ . State the implied comovement between p and GDP y .

Solution. (a) Firm chooses n to maximize $zn^\alpha - wn$, so $w = \alpha zn^{\alpha-1}$. Wage bill $wn = \alpha zn^\alpha = \alpha y$. Hence $d = y - wn = (1 - \alpha)y$.

(b) Household intratemporal FOC gives $w/c = \psi \Rightarrow w = \psi c$. With $c = y = zn^\alpha$, set $\psi y = \alpha zn^{\alpha-1} \Rightarrow \psi zn^\alpha = \alpha zn^{\alpha-1} \Rightarrow \psi n = \alpha \Rightarrow n = \alpha/\psi$. Then $y = z(\alpha/\psi)^\alpha$.

(c) From (S) and using $c = y$, $d' = (1 - \alpha)y'$:

$$\frac{p}{y} = \beta \mathbb{E}_{z'|z} \left[\frac{p'}{y'} + (1 - \alpha) \right] = \beta(1 - \alpha) + \beta \mathbb{E}_{z'|z} \left[\frac{p'}{y'} \right].$$

If $p/y = \Lambda$ constant, $\Lambda = \beta(1 - \alpha) + \beta\Lambda \Rightarrow \Lambda = \frac{\beta(1-\alpha)}{1-\beta}$. Thus $p = \Lambda y$: stock price is procyclical and moves one-for-one with GDP in percentage terms (constant price-output ratio).

Question 3 (RBC: facts/HP filter, BGP, deflation, measurement) [30

pts]

This question uses the notation from the RBC Topic 1A and Topic 1BC slides.

(a) **HP filter (objective and notation).** [12 pts]

Given data $\{y_t\}_{t=1}^T$, the HP filter chooses a trend $\{\tau_t\}_{t=1}^T$. Write the HP minimization problem exactly as in the slides:

$$\min_{\{\tau_t\}_{t=1}^T} \frac{1}{T} \sum_{t=1}^T (y_t - \tau_t)^2 + \frac{\lambda}{T-1} \sum_{t=2}^{T-1} \left[(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1}) \right]^2,$$

and define the cycle $\tilde{y}_t$. In one sentence, what does λ control?

(b) **Balanced growth logic.** [8 pts]

In the one-sector growth model with labor-augmenting progress,

$$Y_t = zF(K_t, n_t X_t), \quad \frac{X_{t+1}}{X_t} = \gamma_x \geq 1,$$

and define gross growth rates $\gamma_v(t) \equiv v_{t+1}/v_t$. Assume z and n_t are constant on the BGP. Explain briefly why a BGP requires

$$\gamma_y = \gamma_c = \gamma_i = \gamma_k \quad \text{and} \quad \gamma_y = \gamma_k = \gamma_x.$$

(c) **Growth-deflated constraints.** [6 pts]

Define detrended (growth-deflated) variables

$$y_t \equiv \frac{Y_t}{X_t}, \quad c_t \equiv \frac{C_t}{X_t}, \quad i_t \equiv \frac{I_t}{X_t}, \quad k_t \equiv \frac{K_t}{X_t}.$$

Write the growth-deflated resource constraint and capital accumulation constraint as in the slides:

$$y_t \geq c_t + i_t, \quad \gamma_x k_{t+1} \leq (1 - \delta)k_t + i_t.$$

(d) **Solow residual / TFP measurement.**

[4 pts]

With Cobb–Douglas production,

$$Y_t = A_t K_t^\alpha (n_t X_t)^{1-\alpha},$$

show that

$$\log A_t = \log Y_t - \alpha \log K_t - (1 - \alpha) \log n_t - (1 - \alpha) \log X_t.$$

Then compute A_t for: $Y_t = 100$, $K_t = 300$, $n_t = 0.30$, $X_t = 2$, $\alpha = 0.34$.

Solution. (a) Cycle is $\tilde{y}_t \equiv y_t - \tau_t$. Larger λ enforces a smoother trend (penalizes changes in trend growth more).

(b) Resource constraint on BGP: $Y_t = C_t + I_t$ with each term growing at constant rates implies $\gamma_y = \gamma_c = \gamma_i$. Capital accumulation $K_{t+1} = (1 - \delta)K_t + I_t$ requires $\gamma_k = \gamma_i = \gamma_y$. CRS production implies $Y_t/X_t = zF(K_t/X_t, n_t)$. With constant z and n_t , constant Y_t/X_t requires K_t/X_t constant, so $\gamma_k = \gamma_x$ and thus $\gamma_y = \gamma_k = \gamma_x$.

(c) Divide $Y_t \geq C_t + I_t$ by X_t to get $y_t \geq c_t + i_t$. Divide $K_{t+1} \leq (1 - \delta)K_t + I_t$ by $X_{t+1} = \gamma_x X_t$ to get $k_{t+1} \leq [(1 - \delta)k_t + i_t]/\gamma_x$, equivalently $\gamma_x k_{t+1} \leq (1 - \delta)k_t + i_t$.

(d) Compute $\log A_t$ then exponentiate. Here: $\log A = \log 100 - 0.34 \log 300 - (0.66) \log 0.30 - (0.66) \log 2$. Numerically: $\log 100 \approx 4.6052$; $\log 300 \approx 5.7038$; $\log 0.30 \approx -1.2040$; $\log 2 \approx 0.6931$. So

$$\log A \approx 4.6052 - 0.34(5.7038) - 0.66(-1.2040) - 0.66(0.6931) \approx 4.6052 - 1.9393 + 0.7946 - 0.4574 \approx 3.0031.$$

Thus $A \approx e^{3.0031} \approx 20.2$.